Bijective Counting, Stars & Bars, PIE & Pigeonhole

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Bijective Counting, Stars & Bars, PIE & Pigeonhole Practice Exercises

1 How many ordered 3-tuples of integers $(x_1, \ldots, x_3)$ satisfy $x_1 + \ldots + x_3 = 9$ subject to $x_i \geq 1$ for every $i = 1, \ldots, 3$?

a. 55

b. 84

c. 28 ✓

d. 25

Shift variables by $y_i = x_i - 1 \geq 0$. Then $y_1 + \ldots + y_3 = 9 - 3(1) = 6$. By Stars and Bars, the number of non-negative integer solutions is $\binom{6+3-1}{3-1} = \binom{8}{2} = 28$.

2 How many ordered 4-tuples of integers $(x_1, \ldots, x_4)$ satisfy $x_1 + \ldots + x_4 = 15$ subject to $x_i \geq 2$ for every $i = 1, \ldots, 4$?

a. 816

b. 120 ✓

c. 330

d. 116

Shift variables by $y_i = x_i - 2 \geq 0$. Then $y_1 + \ldots + y_4 = 15 - 4(2) = 7$. By Stars and Bars, the number of non-negative integer solutions is $\binom{7+4-1}{4-1} = \binom{10}{3} = 120$.

3 How many ordered 5-tuples of integers $(x_1, \ldots, x_5)$ satisfy $x_1 + \ldots + x_5 = 13$ subject to $x_i \geq 1$ for every $i = 1, \ldots, 5$?

a. 495 ✓

b. 2380

c. 1287

d. 490

Shift variables by $y_i = x_i - 1 \geq 0$. Then $y_1 + \ldots + y_5 = 13 - 5(1) = 8$. By Stars and Bars, the number of non-negative integer solutions is $\binom{8+5-1}{5-1} = \binom{12}{4} = 495$.

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4 How many ordered 3-tuples of integers $(x_1, \ldots, x_3)$ satisfy $x_1 + \ldots + x_3 = 15$ subject to $x_i \geq 2$ for every $i = 1, \ldots, 3$?

a. 136

b. 220

c. 52

d. 55 ✓

Shift variables by $y_i = x_i - 2 \geq 0$. Then $y_1 + \ldots + y_3 = 15 - 3(2) = 9$. By Stars and Bars, the number of non-negative integer solutions is $\binom{9+3-1}{3-1} = \binom{11}{2} = 55$.

5 How many integers in $\{1, 2, \ldots, 180\}$ are divisible by at least one of 2, 3, or 5?

a. 48

b. 138

c. 132 ✓

d. 128

By 3-set Inclusion-Exclusion, $|A \cup B \cup C| = \lfloor 180/2 \rfloor + \lfloor 180/3 \rfloor + \lfloor 180/5 \rfloor - \lfloor 180/6 \rfloor - \lfloor 180/10 \rfloor - \lfloor 180/15 \rfloor + \lfloor 180/30 \rfloor = 132$.

6 How many integers in $\{1, 2, \ldots, 210\}$ are divisible by at least one of 2, 3, or 7?

a. 60

b. 150 ✓

c. 155

d. 146

By 3-set Inclusion-Exclusion, $|A \cup B \cup C| = \lfloor 210/2 \rfloor + \lfloor 210/3 \rfloor + \lfloor 210/7 \rfloor - \lfloor 210/6 \rfloor - \lfloor 210/14 \rfloor - \lfloor 210/21 \rfloor + \lfloor 210/42 \rfloor = 150$.

7 How many integers in $\{1, 2, \ldots, 280\}$ are divisible by at least one of 2, 5, or 7?

a. 184 ✓

b. 96

c. 188

d. 180

By 3-set Inclusion-Exclusion, $|A \cup B \cup C| = \lfloor 280/2 \rfloor + \lfloor 280/5 \rfloor + \lfloor 280/7 \rfloor - \lfloor 280/10 \rfloor - \lfloor 280/14 \rfloor - \lfloor 280/35 \rfloor + \lfloor 280/70 \rfloor = 184$.

8 How many integers in $\{1, 2, \ldots, 315\}$ are divisible by at least one of 3, 5, or 7?

a. 144

b. 174

c. 167

d. 171 ✓

By 3-set Inclusion-Exclusion, $|A \cup B \cup C| = \lfloor 315/3 \rfloor + \lfloor 315/5 \rfloor + \lfloor 315/7 \rfloor - \lfloor 315/15 \rfloor - \lfloor 315/21 \rfloor - \lfloor 315/35 \rfloor + \lfloor 315/105 \rfloor = 171$.

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9 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 6 beads can be formed using up to 2 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 11

b. 16

c. 14 ✓

d. 12

Under the cyclic rotation group C_6, rotation by k steps decomposes into $\gcd(6, k)$ cycles, fixing $2^{\gcd(6,\,k)}$ colorings. Grouping by $d = \gcd(6, k) \mid 6$ via Burnside's Lemma gives $\frac{1}{6}\sum_{d \mid 6} \varphi(6/d) \cdot 2^d = \frac{84}{6} = 14$.

10 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 6 beads can be formed using up to 3 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 122

b. 130 ✓

c. 133

d. 127

Under the cyclic rotation group C_6, rotation by k steps decomposes into $\gcd(6, k)$ cycles, fixing $3^{\gcd(6,\,k)}$ colorings. Grouping by $d = \gcd(6, k) \mid 6$ via Burnside's Lemma gives $\frac{1}{6}\sum_{d \mid 6} \varphi(6/d) \cdot 3^d = \frac{780}{6} = 130$.

11 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 8 beads can be formed using up to 2 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 36 ✓

b. 32

c. 38

d. 34

Under the cyclic rotation group C_8, rotation by k steps decomposes into $\gcd(8, k)$ cycles, fixing $2^{\gcd(8,\,k)}$ colorings. Grouping by $d = \gcd(8, k) \mid 8$ via Burnside's Lemma gives $\frac{1}{8}\sum_{d \mid 8} \varphi(8/d) \cdot 2^d = \frac{288}{8} = 36$.

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12 Using Burnside's Orbit-Counting Lemma, how many rotationally distinct necklaces of 9 beads can be formed using up to 2 colors (where two necklaces are identical if one is a rotation of the other, without flipping)?

a. 57

b. 62

c. 58

d. 60 ✓

Under the cyclic rotation group C_9, rotation by k steps decomposes into $\gcd(9, k)$ cycles, fixing $2^{\gcd(9,\,k)}$ colorings. Grouping by $d = \gcd(9, k) \mid 9$ via Burnside's Lemma gives $\frac{1}{9}\sum_{d \mid 9} \varphi(9/d) \cdot 2^d = \frac{540}{9} = 60$.

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